Submission1

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My choices

- Gene: GNLY
- Continuous covariate: mat_vitD (vitamin D, ng/mL)
- Categorical covariate 1: mat_BMI (BMI group, derived from a continuous variable)
- Categorical covariate 2: preg_preeclampsia (Normal vs. Preeclampsia)

Step 1. Load the raw data

```
#read.csv with row.names = 1 uses the first column (probe IDs) as row names
covar <- read.csv(file = 'GSE85307_covar-1.csv')
geneNames <- read.csv(file = 'GSE85307_geneNames.csv')
expr <- read.csv(file = 'GSE85307_geneExpression.csv', row.names = 1)

# The first column name of the covariate file can come in with a hidden extra
# character (BOM), so we rename it ourselves
colnames(covar)[1] <- 'subjectID'

# Look at the the structure
str(covar)
```

```
## 'data.frame':    157 obs. of  10 variables:
##  $ subjectID                   : chr  "6053887-M" "3064425-M" "6094898-M" "7050617-M" ...
##  $ geoID                       : chr  "GSM2264346" "GSM2264347" "GSM2264348" "GSM2264349" ...
##  $ mat_age_yrs                 : chr  "age_years: 31.06639288" "age_years: 27.79466119" "age_years: 3
##  $ preg_preeclampsia           : chr  "pregnancy_condition: Preeclampsia" "pregnancy_condition: Norma
##  $ delivery_gestAge_wks        : chr  "gestation_weeks: 37.42857143" "gestation_weeks: 39.42857143"
##  $ sampleCollection_gestAge_wks: chr  "blood_draw_at_gestation_week: 10" "blood_draw_at_gestation_we
##  $ batch_Microarray            : chr  "microarray_hybridization_date: 03/15/2013" "microarray_hybridi
##  $ mat_BMI                     : chr  "bmi_kg/m2: 37.52407538" "bmi_kg/m2: 20.59144273" "bmi_kg/m2: 1
##  $ mat_vitD                    : chr  "vitamind_25(0h)_ng/ml_whole_blood_baseline: 7.94" "vitamind_25
##  $ mat_asthma                  : chr  "asthma_0=healthy_1=asthmatic: 1" "asthma_0=healthy_1=asthmatic
```

```
dim(expr)
```

```
## [1] 500 157
```

```
head(geneNames)
```

```
##   PROBEID      GENE
```

```
## 1 8128043      CNR1
## 2 7962489 PLEKHA8P1
## 3 7920271    S100A4
## 4 7900468      NFYC
## 5 8026496 LINC00661
## 6 7989725     PDCD7
```

Step 2. Clean the covariates

Every covariate value is stored as text with a label, like "age_year: xxx". I remove everything up to and including ":" and then fix the data types.

```
# Loop through columns 3 to 10 (all covariates; columns 1-2 are the IDs)
# '.*: ' means "any characters followed by a colon and a space"
for(i in colnames(covar)[3:10]){
  covar[,i] <- sub(pattern = '.*: ',replacement = '', x = covar[,i])
}
head(covar)
```

```
##   subjectID      geoID mat_age_yrs preg_preeclampsia delivery_gestAge_wks
## 1 6053887-M GSM2264346 31.06639288      Preeclampsia          37.42857143
## 2 3064425-M GSM2264347 27.79466119            Normal          39.42857143
## 3 6094898-M GSM2264348 34.31895962            Normal          38.85714286
## 4 7050617-M GSM2264349 33.42642026            Normal          39.14285714
## 5 7016015-M GSM2264350 28.27926078      Preeclampsia          40.14285714
## 6 7054360-M GSM2264351 23.50171116            Normal          41.14285714
##   sampleCollection_gestAge_wks batch_Microarray     mat_BMI mat_vitD mat_asthma
## 1                           10       03/15/2013 37.52407538     7.94          1
## 2                  17.42857143       03/15/2013 20.59144273     40.3          1
## 3                           13       03/15/2013 20.71118512     40.8          1
## 4                  17.28571429       03/15/2013 31.99358316     52.2          0
## 5                  10.71428571       03/15/2013 27.17734484     22.8          0
## 6                  10.57142857       03/19/2013 22.28684666     44.6          0
```

```
# Numeric variables
covar$mat_age_yrs <- as.numeric(covar$mat_age_yrs)
covar$delivery_gestAge_wks <- as.numeric(covar$delivery_gestAge_wks)
covar$sampleCollection_gestAge_wks <- as.numeric(covar$sampleCollection_gestAge_wks)
covar$mat_BMI <- as.numeric(covar$mat_BMI)
covar$mat_vitD <- as.numeric(covar$mat_vitD)

# Categorical variables
covar$preg_preeclampsia <- factor(covar$preg_preeclampsia,
                                  levels = c('Normal','Preeclampsia'))
covar$mat_asthma <- factor(ifelse(covar$mat_asthma=='1','Asthma','Healthy'))

# Check that the types and values look right
str(covar)
```

```
## 'data.frame':    157 obs. of  10 variables:
##  $ subjectID                  : chr  "6053887-M" "3064425-M" "6094898-M" "7050617-M" ...
##  $ geoID                      : chr  "GSM2264346" "GSM2264347" "GSM2264348" "GSM2264349" ...
```

```
##  $ mat_age_yrs                 : num  31.1 27.8 34.3 33.4 28.3 ...
##  $ preg_preeclampsia           : Factor w/ 2 levels "Normal","Preeclampsia": 2 1 1 1 2 1 1 1 1 1 ...
##  $ delivery_gestAge_wks        : num  37.4 39.4 38.9 39.1 40.1 ...
##  $ sampleCollection_gestAge_wks: num  10 17.4 13 17.3 10.7 ...
##  $ batch_Microarray            : chr  "03/15/2013" "03/15/2013" "03/15/2013" "03/15/2013" ...
##  $ mat_BMI                     : num  37.5 20.6 20.7 32 27.2 ...
##  $ mat_vitD                    : num  7.94 40.3 40.8 52.2 22.8 44.6 26.2 21.8 39.2 42.9 ...
##  $ mat_asthma                  : Factor w/ 2 levels "Asthma","Healthy": 1 1 1 2 2 2 1 1 2 1 ...
```

```
summary(covar)
```

```
##      subjectID         geoID      mat_age_yrs       preg_preeclampsia
##  Length   :157   Length   :157   Min.   :18.43   Normal      :110
##  N.unique :157   N.unique :157   1st Qu.:22.15   Preeclampsia: 47
##  N.blank  :  0   N.blank  :  0   Median :26.57
##  Min.nchar:  9   Min.nchar: 10   Mean   :26.38
##  Max.nchar:  9   Max.nchar: 10   3rd Qu.:30.14
##                                  Max.   :38.76
##  delivery_gestAge_wks sampleCollection_gestAge_wks  batch_Microarray
##  Min.   :26.43        Min.   :10.00                 Length   :157
##  1st Qu.:38.14        1st Qu.:12.14                 N.unique :  8
##  Median :39.29        Median :13.43                 N.blank  :  0
##  Mean   :38.82        Mean   :14.13                 Min.nchar: 10
##  3rd Qu.:40.00        3rd Qu.:16.43                 Max.nchar: 10
##  Max.   :42.00        Max.   :18.86
##     mat_BMI         mat_vitD       mat_asthma
##  Min.   :17.95   Min.   : 5.11   Asthma :63
##  1st Qu.:22.90   1st Qu.:13.40   Healthy:94
##  Median :27.33   Median :20.40
##  Mean   :28.94   Mean   :22.98
##  3rd Qu.:33.27   3rd Qu.:29.50
##  Max.   :61.95   Max.   :80.80
```

Step 3. Derive a categorical variable from a continuous one

```
# BMI groups: <25, 25-30, >=30
# Only 5 women have the BMI < 18.5, too few for their own group, so they are
# combined with the "<25" group
covar$bmi_group <- cut(covar$mat_BMI,
                       breaks = c(0,25,30,Inf),
                       labels = c('<25','25-30','>=30'))

# Check the sample sizes
table(covar$bmi_group)
```

```
##
##   <25 25-30  >=30
##    53    47    57
```

Step 4. Find my gene and match it to the covariates

The expression columns are in a different order than the rows of the covriate file, so I cannot just paste them together. I sort both by sample ID and then check that they line up.

```
geneOfInterest <- 'GNLY'
# Gene name -> probe ID (the row names of expr are the probe IDs, stored as text)
probeID <- geneNames[which(geneNames$GENE==geneOfInterest),'PROBEID']
probeID <- as.character(probeID)

# Make sure there is exactly one probe and that it is in the expression data
length(probeID)
```

```
## [1] 1
```

```
probeID %in% rownames(expr)
```

```
## [1] TRUE
```

```
# Transpose so samples are rows and pull out the gene's column
exprT <- t(expr)
geneVec <- exprT[,probeID]

# Sort both the covariates and the gene values by sampleID
covar <- covar[order(covar$geoID),]
geneVec <- geneVec[order(names(geneVec))]

# Check every sample must line up(all True)
table(covar$geoID==names(geneVec))
```

```
##
## TRUE
##  157
```

```
# Only add the gene to the data frame
covar$Expression <- as.numeric(geneVec)
plotData <- covar

# Spot check one sample against the raw expression table (should be TRUE)
expr[probeID,plotData$geoID[1]] == plotData$Expression[1]
```

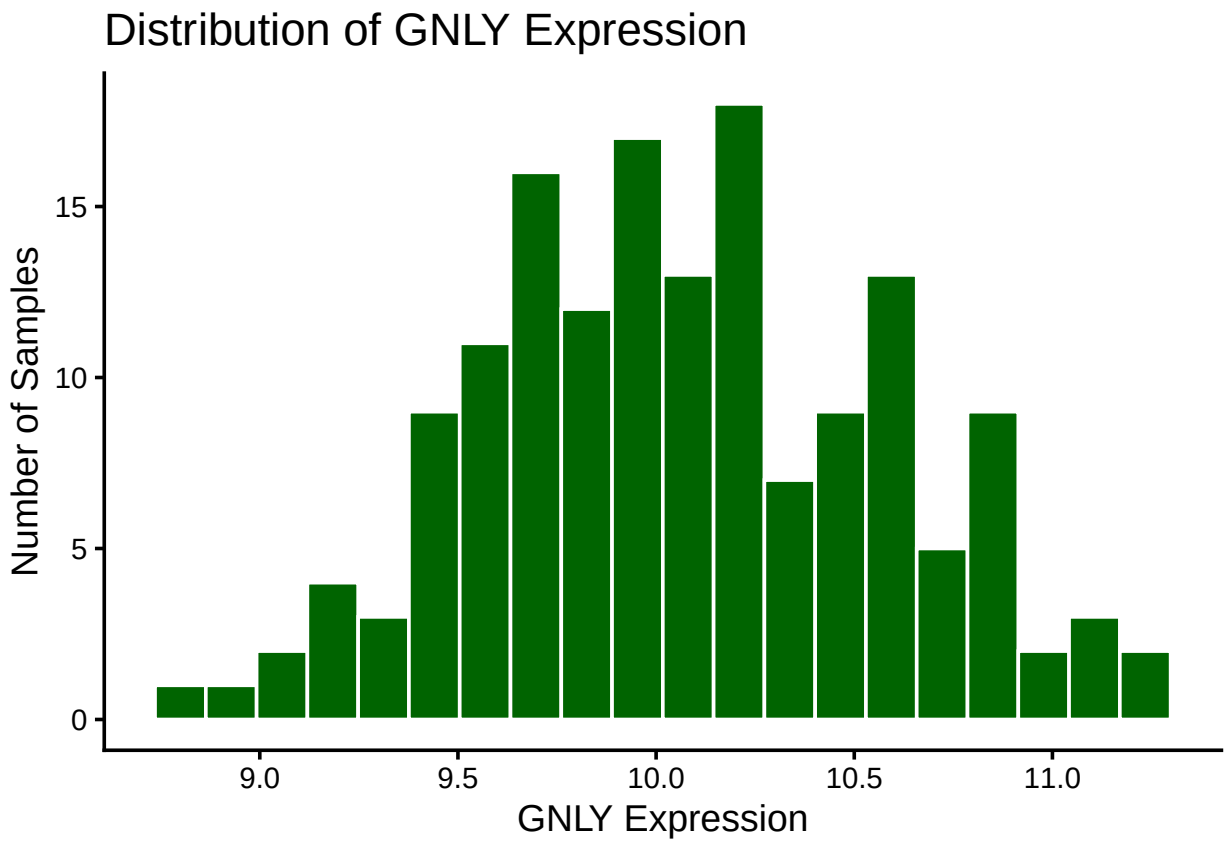
```
## [1] TRUE
```

```
# Check the sample sizes per group
table(plotData$bmi_group,plotData$preg_preeclampsia)
```

```
##
##         Normal Preeclampsia
##   <25       40           13
##   25-30     39            8
##   >=30      31           26
```

Plot 1. Histogram of gene expression

```
ggplot(plotData,aes(x=Expression)) +
  geom_histogram(bins=20,fill='darkgreen',color='white')+
  labs(title = paste('Distribution of',geneOfInterest,'Expression'),
    x=paste(geneOfInterest,'Expression'),
       y='Number of Samples')+
  theme_classic(base_size = 14)
```




```
ggsave('submission1_histogram.png')
```

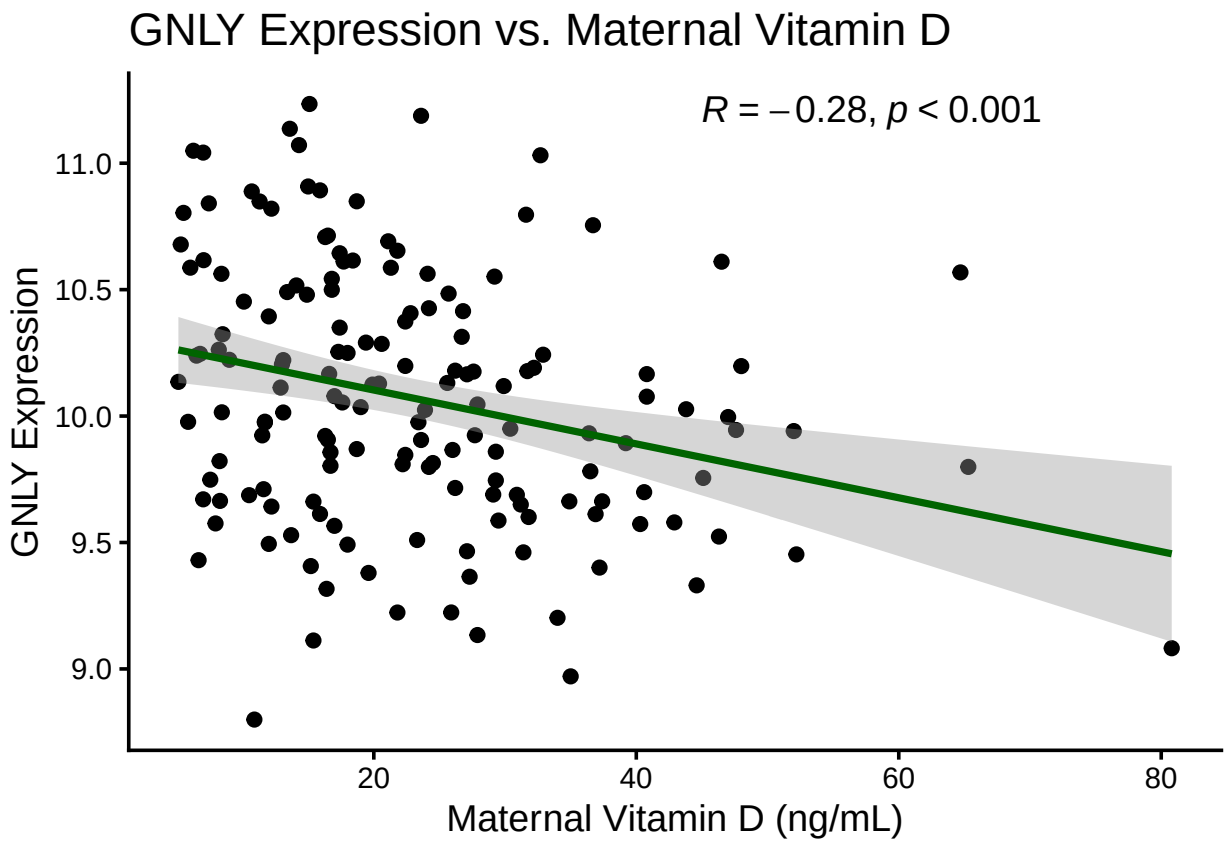
```
## Saving 6.5 x 4.5 in image
```

Plot 2. Continuous covariate vs. gene expression (scatter plot)

```
ggplot(plotData,aes(x=mat_vitD,y=Expression))+
  geom_point(size = 2)+
  geom_smooth(method = lm, color = 'darkgreen')+
  ggpubr::stat_cor(method = 'pearson',
                   label.x = 45,
                   label.y = 11.2,
                   p.accuracy = 0.001)+
```

```
  # label.x / label.y move the label to the empty upper-right corner;
  # p.accuracy = 0.001 stops it from printing a long p-value
  labs(title = paste(geneOfInterest,'Expression vs. Maternal Vitamin D'),
    x='Maternal Vitamin D (ng/mL)',
       y=paste(geneOfInterest,'Expression'))+
  theme_classic(base_size = 14)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
ggsave('submission1_scatter.png')
```

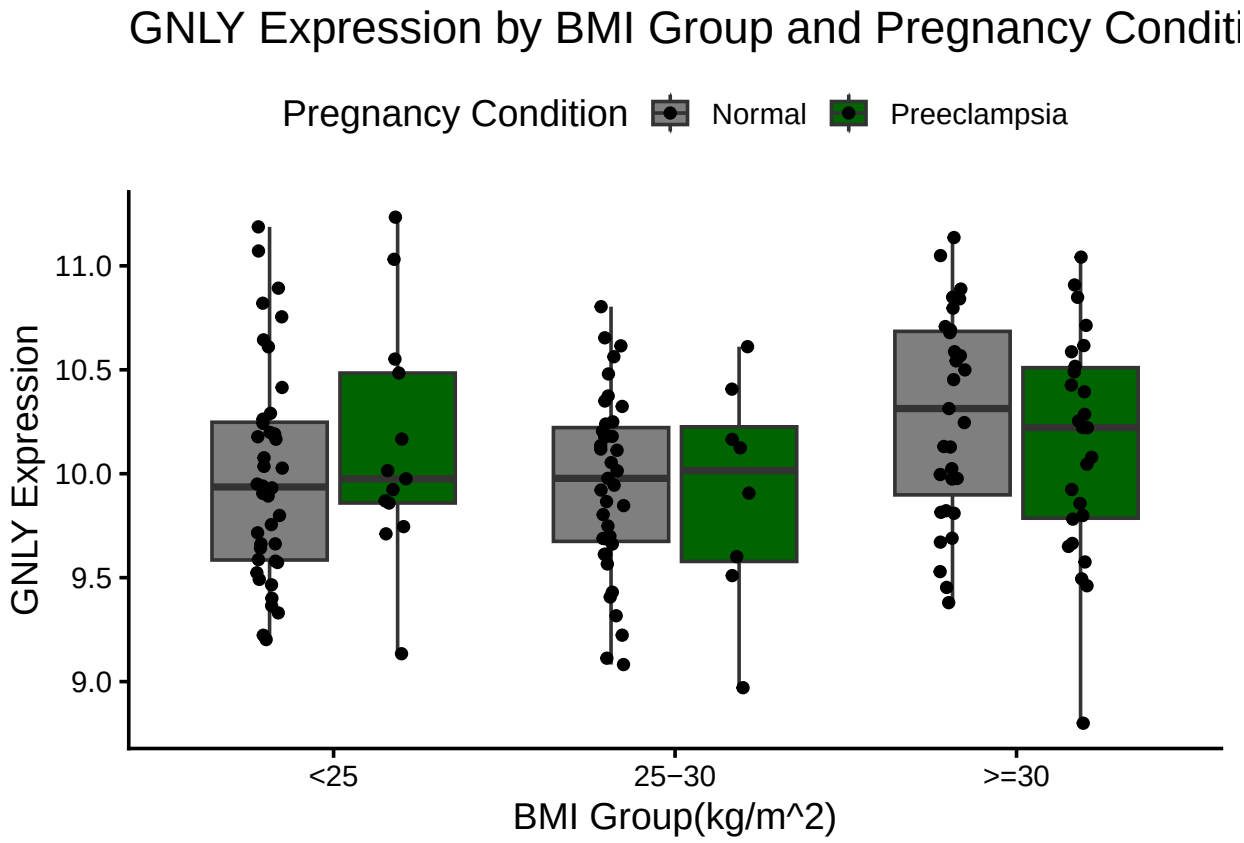
```
## Saving 6.5 x 4.5 in image
## `geom_smooth()` using formula = 'y ~ x'
```

Plot 3. Boxplot by both categorical variables

```
# Define the color palette
colorPalette <- c('grey50','darkgreen')

ggplot(plotData,aes(x=bmi_group,y=Expression,fill=preg_preeclampsia))+
  geom_boxplot(outlier.shape = NA)+
  # outlier.shape = NA hides the boxplot's own outlier dots, because every
```

```
  # sample is drawn as a point below
  geom_point(aes(group=preg_preeclampsia),
             position = position_jitterdodge(jitter.width = 0.15),
             size = 1.5)+
  # Individual samples on top of the boxes. It spreads points inside each box
  # so they do not overlap.
  scale_fill_manual(values = colorPalette)+
  labs(title = paste(geneOfInterest,
                     'Expression by BMI Group and Pregnancy Condition'),
    x='BMI Group(kg/m^2)',
       y=paste(geneOfInterest,'Expression'),
       fill='Pregnancy Condition')+
  theme_classic(base_size = 14)+
  theme(legend.position = 'top')
```




```
ggsave('submission1_boxplot.png')
```

```
## Saving 6.5 x 4.5 in image
```